

Chapter Banking

1. Naveen deposits ₹800 every month in a recurring deposit account for 6 months. If he receives ₹4884 at the time of maturity, then the interest he earns is:

- | | |
|---------|----------|
| (a) ₹84 | (b) ₹42 |
| (c) ₹24 | (d) ₹284 |

Main 2023

2. Manish deposited ₹1500 per month in a recurring deposit account for 2 years. What is the amount payable to him on maturity if the rate of interest is 8% per annum.

- | | |
|------------|------------|
| (a) ₹3,900 | (b) ₹4,200 |
| (c) ₹4,500 | (d) ₹5,000 |

Comp 2023

3. Salman deposits ₹1000 every month in a recurring deposit account for 2 years. If he receives ₹26000 on maturity, find :

- (a) the total interest Salman earns.
(b) the rate of interest.

Main 2023

4. Ram has a recurring deposit of ₹400 per month at 10% per annum. if he gets ₹16620 at the time of maturity, find total time for which the account was held.

Comp 2023

5. Mohit opened a recurring deposit account in a bank for 2 years. He deposit ₹1,000 every month and receives ₹25,500 on maturity. The interest he earned in 2 years is :

(a) ₹13,500

(b) ₹3,000

(c) ₹24,000

(d) ₹1,500

Main 2021 Sem I

6. A man deposit ₹500 per month for 6 months and received ₹3,300 as the maturity value. The interest received by him is :

(a) ₹1,950

(b) ₹300

(c) ₹2,800

(d) None of these

Main 2021 Sem I

Solution

1.

Ans:

Monthly deposit = ₹800

Number of months = 6

Total amount deposited = $800 \times 6 = ₹4800$

Maturity amount = ₹4884

Interest earned = $4884 - 4800 = ₹84$

Correct answer: (a) ₹84

2.

Ans:

Monthly deposit = ₹150

Time = 2 years = 24 months

Rate = 8% p.a.

Total deposit

$$150 \times 24 = ₹360$$

Interest on RD

$$\begin{aligned} I &= \frac{P \times n(n+1)}{2} \times \frac{r}{12 \times 100} \\ &= \frac{150 \times 24 \times 25}{2} \times \frac{8}{1200} = ₹300 \end{aligned}$$

Maturity amount

$$3600 + 300 = ₹3900$$

Correct answer: (a) ₹3,900

3.

Ans:

Given:

- Monthly deposit = ₹1000
- Time = 2 years = 24 months
- Maturity amount = ₹26,000

(a) Total interest earned

Total amount deposited

$$1000 \times 24 = ₹24,000$$

Interest earned

$$26000 - 24000 = ₹2,000$$

(b) Rate of interest

Formula for interest on RD:

$$I = \frac{P \times n(n+1)}{2} \times \frac{r}{12 \times 100}$$

Substitute values:

$$2000 = \frac{1000 \times 24 \times 25}{2} \times \frac{r}{1200}$$

$$2000 = 12500 \times \frac{r}{1200}$$

$$r = \frac{2000 \times 1200}{12500} = 8\%$$

Correct answer:

(a) Total interest = ₹2,000

(b) Rate of interest = 8% per annum

4.

Ans:

Given:

Monthly deposit = ₹400

Rate = 10% p.a.

Maturity amount = ₹16,620

Let number of months = n

Total deposit = $400n$

Interest on RD:

$$I = \frac{400 \times n(n+1)}{2} \times \frac{10}{12 \times 100}$$

Maturity amount:

$$400n + \frac{400 \times n(n+1)}{2} \times \frac{10}{1200} = 16620$$

Solving, we get:

$$n = 36 \text{ months}$$

Correct answer:

Total time = **36 months (3 years)**

5.

Ans:

Monthly deposit = ₹1,000

Time = 2 years = 24 months

Total amount deposited

$$1000 \times 24 = ₹24,000$$

Maturity amount = ₹25,500

Interest earned

$$25,500 - 24,000 = ₹1,500$$

Correct answer:

Interest earned in 2 years = ₹1,500

6.

Ans:

Monthly deposit = ₹500

Time = 6 months

Total amount deposited

$$500 \times 6 = ₹3,000$$

Maturity value = ₹3,300

Interest received

$$3300 - 3000 = ₹300$$

Correct answer: (b) ₹300